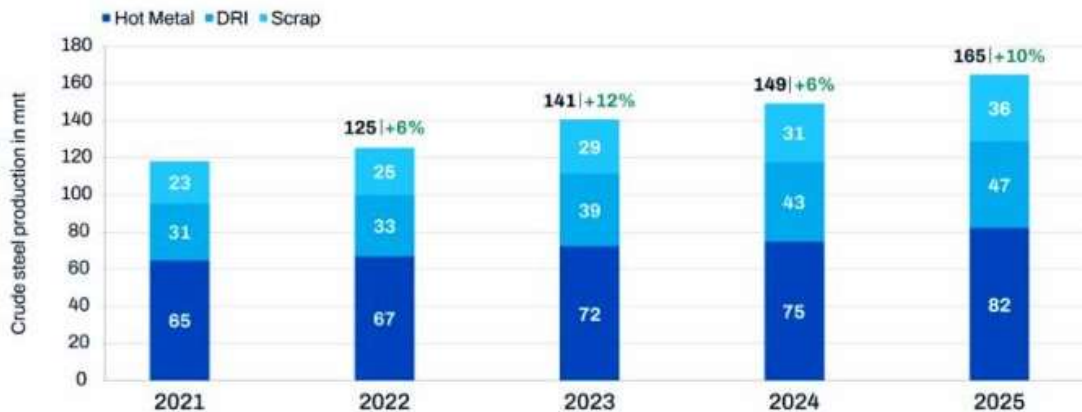


Chapter - 6

Raw Materials for Steel Plants

6.1 India's Metallic Mix for Steel Production



Source: Bigmint

India's crude steel production has grown at ~8–9% CAGR between 2021 and 2025 (119 → 165 mnt), but the growth profile is stabilising — from a sharp 12% surge in 2023 to a more normalized 6–10% range thereafter. This suggests a shift from post-pandemic rebound growth to structurally driven capacity-led expansion.

The metallic intensity pattern is equally telling. Hot metal's share has remained broadly stable (~54–55%), indicating that the BF-BOF route continues to anchor India's steelmaking backbone despite decarbonisation narratives. This reflects ongoing brownfield expansions by integrated players rather than a structural shift away from coal-based steelmaking.

However, the faster growth of DRI and scrap in absolute terms signals diversification. Scrap usage has grown at a higher relative pace (~12% CAGR), improving its share gradually from ~19% to ~22%. This indicates:

- Increasing EAF/IF participation
- Gradual formalisation of scrap supply chains
- Early signs of circularity integration

DRI's steady rise also reflects India's continued comparative advantage in coal-based DRI and emerging gas-based DRI capacity, positioning the sector for potential transition toward green hydrogen in the long term.

Strategically, the data implies:

- India is expanding capacity without fundamentally altering its metallic backbone yet.
- Decarbonisation is incremental, not disruptive.
- Scrap intensity is improving, but structural constraints (scrap availability, quality, logistics) still limit a sharp shift toward EAF dominance.

SAIL has planned a major expansion from its present level of 20 MT Hot Metal in 2024-25 to around 37 MT in 2032-33. One of the major challenges for this expansion is raw materials securitization for ores and fluxes. De-carbonization is a major challenge for steel industry. A major initiative envisaged by SAIL for mitigating carbon footprint is enhanced usage of pellets. Another challenge is use of low grade materials by processes like beneficiation, silica reduction etc. SAIL is poised for expansion in its mines, beneficiation plants, crushing facilities, pellet plants and logistics. SAIL is primarily dependent on procurement to meet its flux requirement. To ensure seamless supply of fluxes, it is imperative to diversify sources.

6.2 Mines of SAIL

The Steel Industry is a raw materials intensive Industry. The importance of raw materials in the steel industry cannot be over emphasized. It affects the Techno-Economics of the Steel Industry. SAIL has the second largest mining outfit in the country spread over the mineral rich states of Jharkhand, Odisha, Chhattisgarh and Madhya Pradesh. The Mines are responsible for supply of consistent quantity and quality of raw materials to steel plants. Specific mission is to achieve rapid expansion and optimisation of its production capacity, improvement in technology and quality and profitability with better service to steel plants. The composite picture of demand of Iron ore is given in Table No. 1

Table – 1: Demand of Iron Ore

Year	Crude Steel Production (mtpa)	Iron Ore Consumption (mtpa)	Linkage of Iron Ore
2024-25	19.17	34.01	Existing Mines
Post Expansion	45 (Capacity 50 MTPA)	78	- SAIL is strengthening its mining capacity along with beneficiation, crushing, pellet plants, and logistics infrastructure. - To meet future requirements, the company plans to scale up iron ore production capacity to about 100 MT by FY 2030-31.

Raw materials like iron ore, limestone, dolomite, manganese ore and coke are used for iron making in steel plants. It is preferable to have low ash content in the coke, low gangue material in the iron ore and fluxes to increase furnace productivity. In order to ensure consistency in quality and quantity of major raw materials SAIL has a number of captive mines. The mines were in the beginning captive to respective steel plants. As our demands grew stringent in terms of quality, it was felt necessary that the mines be brought under one umbrella.

Raw Materials

Iron Ore

MINE	Capacity MTPA	Production 2022-23	Production 2023-24	2024-25	2025-26 (Apr'25- Dec'25)	
Kiriburu	5.5	4.05	3.62	3.81	3.23	
Meghahatuburu	6	3.67	3.24	2.50	1.54	
Gua	4.35	3.96	4.16	4.02	2.96	
Chiria (Manoharpur)	0.75	0.34	0.49	0.41	0.18	
Bolani	10	7.11	7.20	7.41	6.36	
Barsua	3	2.29	2.48	1.85	1.58	
Taldih	2	1.34	1.34	1.76	1.28	
Kalta	4	3.19	3.19	3.37	2.42	
Rajhara Mechanical	8.6	3.54	3.68	6.93	4.71	
Mahamaya						
Dulki						
Jharandalli		4.22	4.37			
Dalli Mechanical						
Kalwar		-	0.02	0.71	0.44	
Rowghat	8	0.07	0.46	1.01	1.23	
TOTAL		33.78	34.34	33.78	25.93	

SAIL Mines Benefits:

- Strategic and need based networking
- Proper development of mines and optimum utilization of assets.
- Create synergy in mining operations.
- Reduction in dependence on non-captive sources by increasing asset utilization/ productivity and inter-plant transfers.
- Perspective planning and development of new sources.
- Centralized responsibility to organize supply of raw materials to plants of desired quality and quantity.
- Proper emphasis on mine planning and long-term development of the mines.
- Integrated approach for development of new mines and scouting for new mineral areas.
- Gainful utilization of surplus resources of mine in others

6.3 Iron ore Mines

Consistent and improved quality of iron ore is crucial for achieving desired blast furnace performance levels. The current production from the operational

iron ore mines have to be augmented to meet SAIL's future requirements for which substantial development of mines will have to be taken up. Presently, SAIL manages 10 iron ore mines and 3 operating flux mines.

Mines	State	Year of Commissioning
Iron Ore		
Kiriburu Iron Ore Mine	Jharkhand	1964
Meghahatuburu Iron Ore Mine	Jharkhand	1985
Bolani Ore Mines	Odisha	1960
Gua Ore Mines	Jharkhand	1919
Manoharpur Ore Mine	Jharkhand	1901
Barsua Iron Mines	Odisha	1960
Kalta Iron Mine	Odisha	1966
Taldih Iron Mine	Odisha	2016
Rajhara, Dalli	Chhattisgarh	
Rowghat	Chhattisgarh	2021

To meet the enhanced iron ore requirement, the following strategies are being planned to be adopted :

- Development of new blocks
- Increased production from existing mines to their potential
- Improving the quality of iron ore by suitable beneficiation
- Achieving operating efficiencies by economic scale of operations

The major linkage for raw materials are in keeping with economics of transportation though it will be pertinent to mention that in order to provide smooth supply of raw materials to Steel Plants there is no specific source as such and raw material is provided as and when required from source which is best suited at that juncture.

SAIL is however, not self-sufficient in fluxes due to poor quality and high cost. Only Kuteshwar Limestone mines are suitable for meeting the requirement for iron making purpose. The mine is being augmented to a higher capacity.

6.4 Coal Mines

Coking coal is a critical input for steelmaking, and due to limited domestic availability, SAIL meets about 80-85% of its requirement through imports, blending it with coal from its own mines and sourced from BCCL/CCL (Coal India Limited).

SAIL operates three coal mines:

- Chasnalla & Tasra (Jharia, Jharkhand) – produce coking coal
- Ramnagore (Raniganj, WB) – produces non-coking (thermal) coal

Raw coking coal from Chasnalla, Tasra and Indigenous Sources (i.e. BBCL) is washed at Chasnalla Washery, which produces about 0.4 MTPA washed coking coal (18% ash) for

SAIL steel plants during FY 24-25. Middlings were also produced which is used for power generation. Washed coking coal is also sourced from BCCL/CCL washeries. Thermal coal from Ramnagore is supplied to steel plants through Chasnalla siding after blending with middlings.

Moreover, SAIL is developing the Tasra Coal Block of 4 MTPA and a 3.5 MTPA Coking Coal Washery through the MDO.

Production figures of Collieries for last 3 years & current year (Apr'25-Dec'25):

(Unit: '000 Tonnes)

Financial Year	2022-23	2023-24	2024-25	2025-26 (Apr'25-Dec'25)
RAW COKING COAL				
Chasnalla	61	104	86	35
Jitpur	8	27	4	-
Tasra	303	334	498	457
Total Raw Coking Coal	372	465	588	492
CLEAN COKING COAL				
Chasnalla Washery	619	484	376	230
THERMAL COAL(Including Jhama& Middlings)				
Ramnagore Incl. Jhama	16	75	98	34
Middlings	553	644	737	513
Total Thermal Coal	569	719	835	547

6.5 Flux Mines

SAIL is currently operating two limestone mines—one at Kuteshwar in Madhya Pradesh and the other at Nandini in Chhattisgarh. In addition, a dolomite mine located at Hirri in Chhattisgarh is also in operation.

Production details of operational flux mines are provided below:

Quantity: Million Tonnes

Sl	Mines	Year of Commissioning	Capacity MTPA	2022-23	2024-25	2025-26 (Apr'25-Dec'25)
1	Kuteshwar Limestone	1974	1.4	1.06	1.10	0.50
2	Nandini Limestone	1960 & 1968	0.83	0.30	0.21	0.15
LIMESTONE TOTAL			2.23	1.36	1.31	0.65
1	Hirri Dolomite	1959	0.94	0.45	0.40	0.44
DOLOMITE TOTAL			0.94	0.45	0.40	0.44
SAIL FLUX			3.17	1.81	1.71	1.09

6.6 Mines Expansion Plan

Aligned with the objectives of the National Steel Policy (NSP) 2017, which envisages achieving 300 MTPA steel capacity by FY 2030-31, SAIL has drawn up a phased plan to enhance its hot metal capacity from about 20 MT in FY 2024-25 to nearly 37 MT. Achieving this growth hinges on long-term raw material security, particularly iron ore, and increased adoption of beneficiation and pelletisation for low-grade ores. Accordingly, SAIL is strengthening its mining capacity along with beneficiation, crushing, pellet plants, and logistics infrastructure. Against an installed iron ore capacity of 46 MTPA, production stood at around 34 MTPA in FY 2024-25 to support crude steel output of 19.17 MT. To meet future requirements, the company plans to scale up iron ore production capacity to about 100 MT by FY 2030-31.
